

The original conductor detects a full inverse-power space

Split-Zero research programme

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The late-heat class in LS1 is $-Q_k(0)[y^{-1}]$. LS13–17 detects that single class. Here we prove that the unchanged conductor detects an explicitly growing space of inverse powers. We give a left inverse using the original lower roots, then carry its bound through the actual minimum metrics. Every conductor coefficient, root difference and physical factor i remains.

1 Original objects and the exact rational response

Keep the simple-quartet domain and complete conductor of KF1–10 and OR1–4, reproduced here where used. Fix odd $k \geq 9$, $k \equiv 1 \pmod{4}$, $0 < \delta < 1/2$, $\gamma > 2$ and the unchanged original period. Set

$$c = k/2, \quad c' = c - 4, \quad q = (k+1)^2, \quad q' = (k-7)^2, \quad D_0 = \sqrt{\delta^2 + \gamma^2}.$$

The roots in the original S coordinate are

$$\beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad 0 \leq a, b \leq k,$$

and the lower roots are

$$\eta_{ab} = c' + (2a - k + 8)\delta + i(2b - k + 8)\gamma, \quad 0 \leq a, b \leq k - 8. \quad (\text{IK1})$$

The monic polynomials χ, χ' have precisely these roots; the prime on χ' denotes the lower polynomial. On $E = \mathbb{C}[S]/(\chi)$ put $y = (S - c)/i$. Every upper y -root is nonzero, so multiplication by y is invertible. Let $\mathcal{J}$ consist of the nonzero actual conductor coefficients $a_j = a_{rs}$, and put

$$J = |\mathcal{J}|, \quad b_j = 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad E_A(z) = \sum_{j \in \mathcal{J}} a_j e^{b_j z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0). \quad (\text{IK2})$$

The shifts are distinct, $\mu_v \neq 0$, and $v \leq J - 1$. Indeed if the first J moments vanished, the Vandermonde matrix on the distinct b_j would kill the nonzero vector (a_j) .

Define the original conductor value map

$$C_A : E \longrightarrow \mathbb{C}^{q'}, \quad (C_A[p])_\eta = \sum_{j \in \mathcal{J}} a_j p(\eta + b_j). \quad (\text{IK3})$$

Root addition in IK1–2 puts each $\eta + b_j$ among the upper roots; therefore IK3 is independent of the representative modulo χ . The complete original kernel satisfies $K \subset \ker C_A$; its further invariant columns remain as in KF2 and MR1–10.

For an integer $s \geq 1$, put $U_s d = \sum_{n=1}^s d_n [y^{-n}]$, where $d \in \mathbb{C}^s$. The exact response is

$$(C_A U_s d)_\eta = R_d(\eta - c), \quad R_d(z) = \sum_{j \in \mathcal{J}} \sum_{n=1}^s \frac{a_j i^n d_n}{(z + b_j)^n}. \quad (\text{IK4})$$

The factor i^n comes from $y^{-n} = i^n (S - c)^{-n}$. No quotient polynomial or coordinate has been replaced.

2 Degree bound and injectivity on the inverse-power space

Let $B_A(z) = \prod_j (z + b_j)$. The function $P_d = B_A^s R_d$ is a polynomial. If $d \neq 0$, write $n_0 = \min\{n : d_n \neq 0\}$ and $t = \max\{n : d_n \neq 0\}$. At every $z = -b_j$, the rational function R_d has a pole of exactly order t , whose highest coefficient is $a_j i^t d_t \neq 0$. At infinity the binomial expansion gives

$$R_d(z) = i^{n_0} d_{n_0} (-1)^v \binom{n_0 + v - 1}{v} \mu_v z^{-n_0 - v} + O(z^{-n_0 - v - 1}). \quad (\text{IK5})$$

To verify the leading term, use $(z + b)^{-n} = \sum_{r \geq 0} (-1)^r \binom{n+r-1}{r} b^r z^{-n-r}$. All moments with $r < v$ vanish. Among terms with exponent $n_0 + v$, each $n > n_0$ has $r < v$ and vanishes too. The displayed coefficient is nonzero. Hence

$$\deg P_d = Js - v - n_0 \leq D_s := Js - v - 1. \quad (\text{IK6})$$

In particular the degree estimate retains every shift.

Assume the explicit finite guard

$$Js - v \leq q'. \quad (\text{IK7})$$

There are q' distinct lower evaluation points $z_\eta = \eta - c$, and no $B_A(z_\eta)$ vanishes. If $C_A U_s d = 0$, then P_d vanishes at all these points. Its degree is at most $D_s < q'$, so $P_d = 0$. The nonzero pole in IK5 then forces $d = 0$. We have proved

$$C_A U_s \text{ is injective, } \quad U_s \mathbb{C}^s \cap K = \{0\}, \quad 1 \leq s \leq \left\lfloor \frac{q' + v}{J} \right\rfloor. \quad (\text{IK8})$$

This also proves independence of these actual inverse powers. It does not replace the full observation kernel by the conductor kernel.

3 A left inverse in original root differences

Select exactly $D_s + 1 = Js - v$ distinct lower roots in lexicographic (a, b) order, and write their centered values as $z_0, \dots, z_{D_s}$. Let C_s be the $(D_s + 1) \times s$ matrix in IK4 restricted to these rows. Define

$$L_\alpha(z) = \prod_{\ell \neq \alpha} \frac{z - z_\ell}{z_\alpha - z_\ell}, \quad P_w(z) = \sum_{\alpha=0}^{D_s} B_A(z_\alpha)^s w_\alpha L_\alpha(z). \quad (\text{IK9})$$

This is Lagrange polynomial interpolation; see Temme [5], with the exact nodal differences in IK9. For $w = C_s d$, the polynomial P_w equals P_d , by the degree bound and its values at all the chosen points.

Fix any $j_* \in \mathcal{J}$ and put $D_*(w) = \prod_{j \neq j_*} (b_j - b_{j_*} + w)^s$. This polynomial is nonzero at zero. In the following coefficient extraction use the scalar indeterminate t , so the input vector w remains distinct. The exact left inverse is

$$(W_s w)_n = \frac{1}{a_{j_*} i^n} [t^{s-n}] \frac{P_w(-b_{j_*} + t)}{D_*(t)}, \quad 1 \leq n \leq s, \quad W_s C_s = I_s. \quad (\text{IK10})$$

To check it without any limit, divide $P_d = B_A^s R_d$ by D_* . At $z = -b_{j_*} + w$ the resulting expression is $w^s R_d$. Its terms of degree below s are exactly $\sum_n a_{j_*} i^n d_n w^{s-n}$. All other summands in IK4 are analytic at this pole and become divisible by w^s . This proves every row of IK10.

For an entirely finite size bound put $R = kD_0$ and $M = s(J - 1)$. Define

$$h_r = \begin{cases} \binom{M+r-1}{r} (2\delta)^{-M-r}, & M > 0, \\ 1, & M = 0, \ r = 0, \\ 0, & M = 0, \ r > 0. \end{cases}$$

The coefficient bound on D_*^{-1} is $|[w^r]D_*^{-1}| \leq h_r$: expand each $(b_j - b_{j_*} + w)^{-s}$ and use $|b_j - b_{j_*}| \geq 2\delta$. Every $|z_\alpha + b_j| \leq R$ is an actual upper-root modulus, and every $|z_\alpha - z_\ell| \geq 2\delta$. Consequently

$$|[w^t]L_\alpha(-b_{j_*} + w)| \leq \binom{D_s}{t} R^{D_s-t} (2\delta)^{-D_s}.$$

Define the explicit nonnegative number

$$\mathcal{W}_s = \frac{\sqrt{s(D_s+1)} R^{Js}}{|a_{j_*}|(2\delta)^{D_s}} \max_{1 \leq n \leq s} \sum_{t=0}^{\min(D_s, s-n)} \binom{D_s}{t} R^{D_s-t} h_{s-n-t}. \quad (\text{IK11})$$

The preceding bounds in IK9–10 give $\|W_s\|_2 \leq \mathcal{W}_s$, by first bounding each entry and then its Frobenius norm. Thus $\|C_s d\|_2 \geq \|d\|_2 / \mathcal{W}_s$. This is a bound in the specified coefficient and literal value coordinates; the next step carries it to the original metrics.

4 Return to the actual attained observation

At any original cutoff N , retain its positive attained metric G_N on E , complete observation $\Lambda : E \rightarrow B$ with kernel K ,

$$Q_N = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad \mathcal{L}_N = G_N^{-1} \Lambda^* Q_N, \quad P_B = \mathcal{L}_N \Lambda.$$

Let $C_{\text{sel}} : E \rightarrow \mathbb{C}^{D_s+1}$ consist of the selected conductor rows. It has full row rank. Indeed the complete C_A has rank q' : $\mathcal{T}_A$ lowers polynomial degree by exactly v with nonzero leading coefficient, so it surjects from degree below q onto degree below $q - v$; this contains the degree-below- q' interpolation space because $q - q' = 16k - 48 \geq 96 > v$. Evaluation at the distinct lower roots is onto. Thus any selected subset of its rows is independent.

Put

$$\Gamma_{\text{sel}} = C_{\text{sel}} G_N^{-1} C_{\text{sel}}^*, \quad H_s = U_s^* G_N U_s, \quad R_N = C_{\text{sel}} \mathcal{L}_N.$$

Both Grams are positive definite. Since C_{sel} kills K , $C_{\text{sel}} = R_N \Lambda$, and direct multiplication gives $R_N Q_N^{-1} R_N^* = \Gamma_{\text{sel}}$. The minimum over $R_N b = w$ of $b^* Q_N b$ is $w^* \Gamma_{\text{sel}}^{-1} w$: its section is $Q_N^{-1} R_N^* \Gamma_{\text{sel}}^{-1}$, and every other lift has an orthogonal nonnegative correction. It follows, in the unchanged coefficient space $\mathbb{C}^s$, that

$$U_s^* \Lambda^* Q_N \Lambda U_s \succeq C_s^* \Gamma_{\text{sel}}^{-1} C_s \succeq \frac{I_s}{\lambda_{\max}(\Gamma_{\text{sel}}) \mathcal{W}_s^2}, \quad (\text{IK12})$$

$$\boxed{\frac{\|P_B U_s d\|_{G_N}^2}{\|U_s d\|_{G_N}^2} \geq \frac{1}{\lambda_{\max}(\Gamma_{\text{sel}}) \mathcal{W}_s^2 \lambda_{\max}(H_s)} > 0 \quad (d \neq 0).}$$

Every matrix here is the original attained matrix or its explicitly proved restriction or quotient. IK12 is a quantitative finite observability statement on a space whose permitted dimension grows as q'/J . It is not a uniform positive limiting fraction: the complete metric factors in its denominator remain. For $s = 1$ it applies to LS1's late-heat line, up to its exact nonzero scalar $-Q_k(0)$. No individual complex phase or kq allocation is assigned by this positive detection theorem.

For a version exposing its coordinate factors, let $G_N^{\text{val}} = V^{-*} G_N V^{-1}$ be the literal upper-root value metric, $\kappa_N^{\text{val}} = \lambda_{\max}(G_N^{\text{val}}) / \lambda_{\min}(G_N^{\text{val}})$ and $A_1 = \sum_j |a_j|$. Each conductor row has Euclidean norm at most A_1 , so the selected matrix has norm at most $\sqrt{D_s + 1} A_1$. Each inverse-power value has modulus at most δ^{-n} , because every upper root has centered real part an odd nonzero multiple of δ . Thus

$$\lambda_{\max}(\Gamma_{\text{sel}}) \leq \frac{(D_s + 1) A_1^2}{\lambda_{\min}(G_N^{\text{val}})}, \quad \lambda_{\max}(H_s) \leq \lambda_{\max}(G_N^{\text{val}}) q \sum_{n=1}^s \delta^{-2n}.$$

Substitution into IK12 gives the explicit original-coordinate bound

$$\frac{\|P_B U_s d\|_{G_N}^2}{\|U_s d\|_{G_N}^2} \geq \left[\kappa_N^{\text{val}} q(D_s + 1) A_1^2 \mathcal{W}_s^2 \sum_{n=1}^s \delta^{-2n} \right]^{-1} > 0. \quad (\text{IK13})$$

For the actual Gamma comparison in KF33–35 and its original full-source transfer, $\log \kappa_N^{\text{val}} = O_h(q \log(q + 2))$. Uniformly over every integer s in IK8, the other factors on the right of IK13 have logarithms at most $O_A(q \log(q + 2))$. Indeed $J_s \leq q' + v \leq q + 80$, $D_s < q'$, and $M = s(J - 1) \leq q + 80$. All binomial indices in IK11 are at most $2q + 160$, so each binomial is bounded by 2^{2q+160} ; there are at most $q + 81$ summands. The powers of $R = kD_0$ have total exponent $O(q)$, while the powers of the fixed 2δ , $|a_{j*}|^{-1}$, and the geometric sum contribute at most $O_A(q)$ in logarithm. This proves a uniform lower bound $\exp[-O_{A,h}(q \log(q + 2))]$ on the entire allowed inverse-power space, at all four original cutoffs. It supplies a finite quantitative receiver for growing inverse powers; its lower bound is allowed to tend to zero.

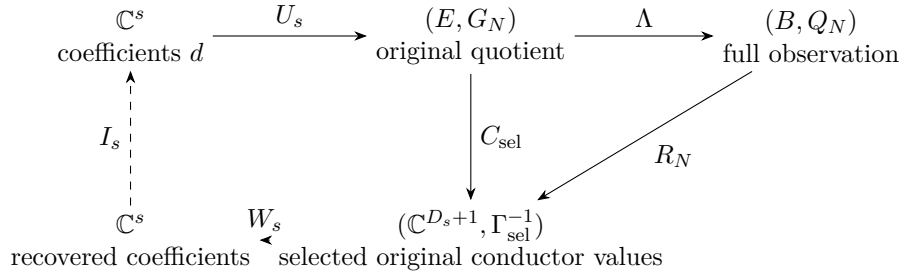


Figure 1: Exact maps IK3–12. The upper triangle satisfies $C_{\text{sel}} = R_N \Lambda$ in the original metrics, and the loop satisfies $W_s C_{\text{sel}} U_s = I_s$. The second identity is the reconstruction in IK9–10, valid when $J_s - v \leq q'$. The positive loss under the first map is the complete minimum proved in IK12. No chosen source norm replaces G_N .

References

- [1] *The actual invariant kernel: degree filtration, quantitative minors, and its complete conductor graph*, KF1–10, complete pinned proof, KF1–10 and KF33–35.
- [2] *The roots of the complete conductor polynomial*, OR1–4, complete pinned proof, OR1–4.
- [3] *The late heat direction through the original observation*, LS1 and LS13–17, 21 September 2026, complete pinned proof.
- [4] Split-Zero research programme, *The original observation kernel as an exact mixed-relation determinant*, MR1–10, complete pinned public proof.
- [5] N. M. Temme, *Numerical Methods*, Chapter 3 of the *NIST Digital Library of Mathematical Functions*, version 1.2.8, 15 September 2026, Section 3.3(i), formulas 3.3.1–2. The original formula TeX has been read and retained. IK9 proves the needed polynomial identity directly on the original complex nodes; no interpolation error theorem is imported.